HR Analytic,s: Employee Attrition Analysis Using Python and SQL

1. Project Overview

This project focuses on analyzing the HR Analytic,s data-set to understand employee attrition patterns, salary distribution, and workforce trends. The primary objective is to clean and processors raw employee data using Python, store the refined data-set in a MySQL database, and perform SOL-based analysis to extract meaningful business insights. The project follows a complete data analytic,s pipeline including data cleaning, exploratory data analysis, database integration, and business intelligence reporting. The insights generated from this analysis can help organizations make data-driven HR decisions to reduce attrition and improve employee retention.

2. data-set Summary

The HR Analytic,s data-set contains employee-level information used to analyze workforce behaviour and attrition trends. The data-set includes both numerical and categorical variables such as Department, JobRole, MonthlyIncome, Attrition, WorkLifeBalance, and YearsWithCurrManager. Initial inspection using Python revealed that the data-set is well structured but contains some missing values, particularly in the YearsWithCurrManager column. Appropriate data cleaning and imputation techniques were applied to handle these missing values. Overall, the data-set is suitable for performing HR Analytic,s and deriving meaningful business insights.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

● **Data Loading:** Imported the HR Analytic,s data-set using pandas into a Data-Frame for further processing.

● **Initial Exploration:** Used `data.info()` to examine the data-set structure and data types, and `data.describe()` to generate summary statistics for numerical features.

```
dtype: int64
        EmpID         Age AgeGroup Attrition BusinessTravel  ...  WorkLifeBalance YearsAtCompany  YearsInCurrentRole  YearsSinceLastPromotion YearsWithCurrManager
count    1480  1480.000000     1480      1480           1480  ...      1480.000000    1480.000000         1480.000000              1480.000000          1423.000000
unique   1470         NaN        5         2              4  ...              NaN            NaN                 NaN                     NaN                 NaN
top     RM1468         NaN    26-35        No  Travel_Rarely  ...              NaN            NaN                 NaN                     NaN                 NaN
freq        2         NaN      611      1242           1042  ...              NaN            NaN                 NaN                     NaN                 NaN
mean      NaN   36.917568      NaN       NaN            NaN  ...         2.760811       7.009459            4.228378                2.182432            4.118060
std       NaN    9.128559      NaN       NaN            NaN  ...         0.707024       6.117945            3.616020                3.219357            3.555484
min       NaN   18.000000      NaN       NaN            NaN  ...         1.000000       0.000000            0.000000                0.000000            0.000000
25%       NaN   30.000000      NaN       NaN            NaN  ...         2.000000       3.000000            2.000000                0.000000            2.000000
50%       NaN   36.000000      NaN       NaN            NaN  ...         3.000000       5.000000            3.000000                1.000000            3.000000
75%       NaN   43.000000      NaN       NaN            NaN  ...         3.000000       9.000000            7.000000                3.000000            7.000000
max       NaN   60.000000      NaN       NaN            NaN  ...         4.000000      40.000000           18.000000               15.000000           17.000000
```

#	Column	Non-Null Count	Dtype
0	EmpID	1480 non-null	object
1	Age	1480 non-null	int64
2	AgeGroup	1480 non-null	object
3	Attrition	1480 non-null	object
4	BusinessTravel	1480 non-null	object
5	DailyRate	1480 non-null	int64
6	Department	1480 non-null	object
7	DistanceFromHome	1480 non-null	int64
8	Education	1480 non-null	int64
9	EducationField	1480 non-null	object
10	EmployeeCount	1480 non-null	int64
11	EmployeeNumber	1480 non-null	int64
12	EnvironmentSatisfaction	1480 non-null	int64
13	Gender	1480 non-null	object
14	HourlyRate	1480 non-null	int64
15	JobInvolvement	1480 non-null	int64
16	JobLevel	1480 non-null	int64
17	JobRole	1480 non-null	object
18	JobSatisfaction	1480 non-null	int64
19	MaritalStatus	1480 non-null	object
20	MonthlyIncome	1480 non-null	int64
21	SalarySlab	1480 non-null	object
22	MonthlyRate	1480 non-null	int64
23	NumCompaniesWorked	1480 non-null	int64
24	Over18	1480 non-null	object
25	OverTime	1480 non-null	object
26	PercentSalaryHike	1480 non-null	int64
27	PerformanceRating	1480 non-null	int64
28	RelationshipSatisfaction	1480 non-null	int64
29	StandardHours	1480 non-null	int64
30	StockOptionLevel	1480 non-null	int64
31	TotalWorkingYears	1480 non-null	int64
32	TrainingTimesLastYear	1480 non-null	int64
33	WorkLifeBalance	1480 non-null	int64
34	YearsAtCompany	1480 non-null	int64
35	YearsInCurrentRole	1480 non-null	int64
36	YearsSinceLastPromotion	1480 non-null	int64
37	YearsWithCurrManager	1423 non-null	float64

● **Missing Data Handling:** Checked for null values across all columns. Missing values in the `YearsWithCurrManager` column were imputed using the median value grouped by `WorkLifeBalance` to maintain distribution consistency.

● **Column Standardization:** Renamed all column names to lowercase (snake-case style) to improve readability, maintain consistency, and ensure SQL compatibility.

● **Data Cleaning:** Removed rows containing excessive null values using appropriate filtering to improve overall data quality.

● **Data Consistency Check:** Verified the data-set for duplicate records and inconsistent entries. Ensured that categorical fields such as `Attrition`, `Department`, and `JobRole` contained standardized values.

● **Database Integration:** Connected the Python script to MySQL using SQLAlchemy and successfully loaded the cleaned Data-Frame into the `Hr_table` for further SQL based business analysis.

4. Data Analysis using SQL (Business Transactions) We performed structured analysis in MYSQL to answer key business questions:

1. **Employee Distribution by Gender** – Compared the total number of employees across male and female categories to understand workforce diversity.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_employees
2903

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2. High-Income Employees – Identified employees whose MonthlyIncome is above the company average to detect top earners.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department	employee_count
Research & Development	1891
Sales	887
Human Resources	125

3.Top 5 Highest Paid Employees – Retrieved the top five employees with the highest monthly income.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

empid	age	agegroup	attrition	businesstravel	dailyrate	department	distancefromhome	education	educationfield	employeecount	employeenumber	environmentsatisfaction	gender
RM297	18	18-25	Yes	Travel_Rarely	230	Research & Development	3	3	Life Sciences	1	405	3	Male
RM458	18	18-25	Yes	Travel_Frequently	1306	Sales	5	3	Marketing	1	614	2	Male
RM829	18	18-25	Yes	Non-Travel	247	Research & Development	8	1	Medical	1	1156	3	Male
RM1154	18	18-25	Yes	Travel_Frequently	544	Sales	3	2	Medical	1	1624	2	Female
RM128	19	18-25	Yes	Travel_Rarely	528	Sales	22	1	Marketing	1	167	4	Male
RM172	19	18-25	Yes	Travel_Frequently	602	Sales	1	1	Technical Degree	1	235	3	Female
RM178	19	18-25	Yes	Travel_Rarely	303	Research & Development	2	3	Life Sciences	1	243	2	Male
RM423	19	18-25	Yes	Travel_Rarely	489	Human Resources	2	2	Technical Degree	1	566	1	Male
RM689	19	18-25	Yes	Travel_Rarely	419	Sales	21	3	Other	1	959	4	Male
RM893	19	18-25	Yes	Non-Travel	504	Research & Development	10	3	Medical	1	1248	1	Female
RM103	20	18-25	Yes	Travel_Frequently	871	Research & Development	6	3	Life Sciences	1	137	4	Female
RM514	20	18-25	Yes	Travel_Rarely	1362	Research & Development	10	1	Medical	1	701	4	Male
RM663	20	18-25	Yes	Travel_Rarely	500	Sales	2	3	Medical	1	922	3	Female
RM690	20	18-25	Yes	Travel_Rarely	129	Research & Development	4	3	Technical Degree	1	960	1	Male
RM732	20	18-25	Yes	Travel_Rarely	1097	Research & Development	11	3	Medical	1	1016	4	Female
RM777	20	18-25	Yes	Travel_Frequently	769	Sales	9	3	Marketing	1	1077	4	Female

4.Work-Life Balance Comparison – Compared average MonthlyIncome across different WorkLifeBalance levels to analyze compensation patterns.

Result Grid | Filter Rows:

JobRole	avg_salary
Laboratory Technician	3243.5205
Sales Representative	2630.1677
Research Scientist	3238.4464
Human Resources	4215.2524
Manufacturing Director	7307.1076
Sales Executive	6942.1159
Healthcare Representative	7530.1351
Research Director	16055.3694
Manager	17163.4703

5.Attrition Analysis – Compared total employee count and attrition count across departments to identify high-risk areas.

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

empid	age	agegroup	attrition	businesstravel	dailyrate	department	distancefromhome	education	educationfield	employeecount	employeenumber	environmentsatisfaction	gender	hourlyrate
RM191	52	46-55	No	Travel_Rarely	699	Research & Development	1	4	Life Sciences	1	259	3	Male	65
RM191	52	46-55	No	Travel_Rarely	699	Research & Development	1	4	Life Sciences	1	259	3	Male	65
RM747	41	36-45	No	Non-Travel	247	Research & Development	7	1	Life Sciences	1	1035	2	Female	55
RM747	41	36-45	No	Non-Travel	247	Research & Development	7	1	Life Sciences	1	1035	2	Female	55
RM852	56	55+	No	Travel_Rarely	718	Research & Development	4	4	Technical Degree	1	1191	4	Female	92

6. Attrition-Prone Job Roles – Identified the top 5 job roles with the highest attrition rate.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

empid	age	agegroup	attrition	businesstravel	dailyrate	department	distancefromhome	education	educationfield	employeecount	employeenumber	environmentsatisfaction	gender
RM139	25	18-25	No	Travel_Rarely	959	Sales	28	3	Life Sciences	1	183	1	Male
RM044	27	26-35	No	Travel_Frequently	994	Sales	8	3	Life Sciences	1	56	4	Male
RM213	27	26-35	No	Travel_Frequently	1242	Sales	20	3	Life Sciences	1	293	4	Female
RM531	27	26-35	No	Travel_Rarely	608	Research & Development	1	2	Life Sciences	1	725	3	Female
RM538	27	26-35	No	Travel_Frequently	294	Research & Development	10	2	Life Sciences	1	733	4	Male
RM555	27	26-35	No	Travel_Rarely	975	Research & Development	7	3	Medical	1	764	4	Female
RM611	27	26-35	No	Travel_Rarely	269	Research & Development	5	1	Technical Degree	1	844	3	Male
RM781	28	26-35	Yes	Non-Travel	1366	Research & Development	24	2	Technical Degree	1	1082	2	Male
RM810	28	26-35	No	Travel_Rarely	950	Research & Development	3	3	Medical	1	1121	4	Female
RM945	28	26-35	No	Non-Travel	1476	Research & Development	1	3	Life Sciences	1	1315	3	Female
RM1042	28	26-35	No	Travel_Rarely	866	Sales	5	3	Medical	1	1469	4	Male
RM1074	28	26-35	No	Travel_Rarely	1083	Research & Development	29	1	Life Sciences	1	1514	3	Male

7. Employee Segmentation by Tenure – Classified employees into New, Mid-Level, and Experienced groups based on TotalWorkingYears.

Result Grid | Filter Rows: | Export: | Wrap Cel

Department	attrition_count
Research & Development	260
Sales	184
Human Resources	24

8. Top Paid Employees per Department – Listed the highest-paid employee within each department.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

empid	age	agegroup	attrition	businesstravel	dailyrate	department	distancefromhome	education	educationfield	employeecount	employeenumber	environmentsatisfaction	gend

9. High Tenure & Attrition Check – Analyzed whether employees with higher YearsAtCompany are less likely to leave the organization.

Result Grid | Filter Rows: | Expo

EmployeeNumber	MonthlyIncome_Clean
405	1420
411	1200
614	1878
1012	1051
1156	1904
1368	1611
1624	1569
1839	1514
167	1675
201	1483
235	2325
243	1102
566	2564

1514

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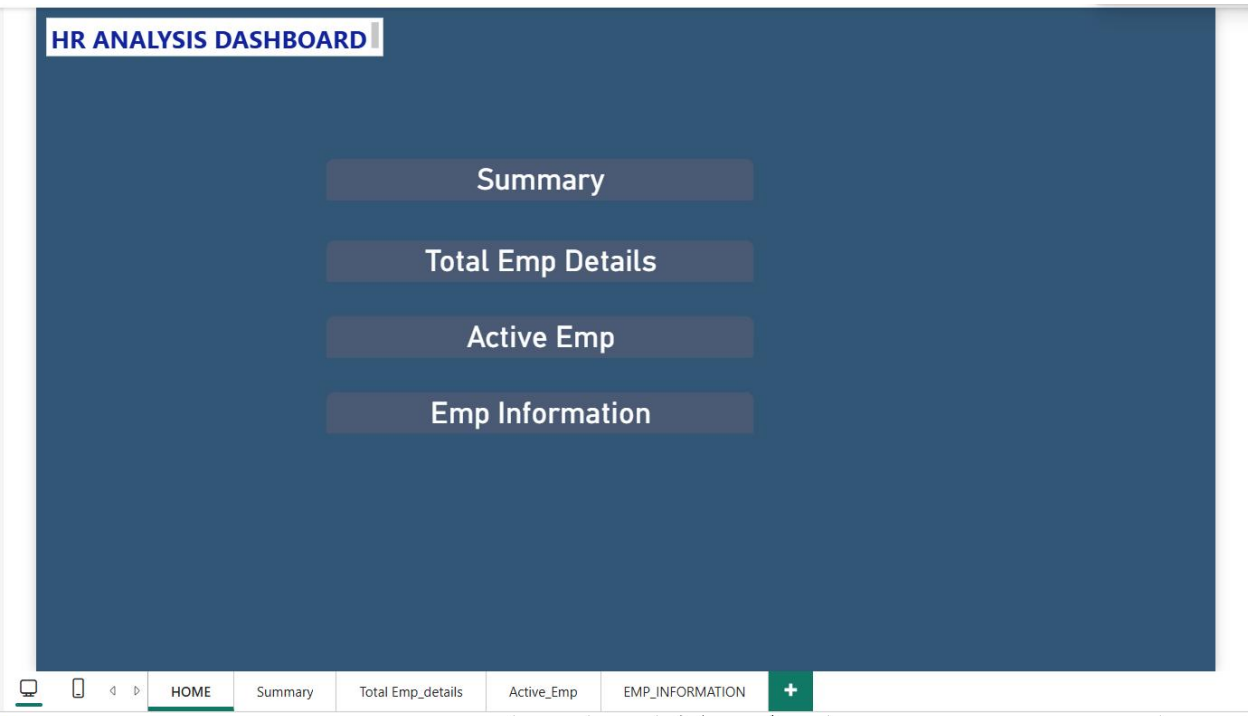
10. Income Distribution by Work-Life Balance – Calculated total and average MonthlyIncome contribution for each WorkLifeBalance category.

empid	age	agegroup	attrition	businesstravel	dailyrate	department	distancefromhome	education	educationfield	employeecount	employeenumber
RM957	56	55+	No	Travel_Rarely	206	Human Resources	8	4	Life Sciences	1	1338
RM957	56	55+	No	Travel_Rarely	206	Human Resources	8	4	Life Sciences	1	1338
RM191	52	46-55	No	Travel_Rarely	699	Research & Development	1	4	Life Sciences	1	259
RM191	52	46-55	No	Travel_Rarely	699	Research & Development	1	4	Life Sciences	1	259
RM919	51	46-55	No	Travel_Frequently	237	Sales	9	3	Life Sciences	1	1282
RM919	51	46-55	No	Travel_Frequently	237	Sales	9	3	Life Sciences	1	1282

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



JobRole	1	2	3	4	Total
Healthcare Representative	18	12	24	27	81
Human Resources	9	10	8	8	35
Laboratory Technician	38	31	49	53	171
Manager	10	13	15	20	58
Manufacturing Director	13	20	28	20	81
Research Director	10	9	16	12	47
Research Scientist	34	33	55	53	175
Total	176	172	262	277	887

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Department	Sum of Education	Sum of EmployeeCount	Sum of EnvironmentSatisfaction	Sum of JobInvolvement	Sum of HourlyRate	Gender	JobRo
Human Resources	95	32	80	87	2191	Male	Human Resourc
Human Resources	15	5	15	14	379	Male	Manager
Research & Development	184	63	166	177	4256	Male	Healthcare Rep
Research & Development	381	140	381	382	9691	Male	Laboratory Tech
Research & Development	58	23	66	61	1582	Male	Manager
Research & Development	175	63	184	170	4069	Male	Manufacturing
Research & Development	131	42	112	115	2587	Male	Research Direct
Research & Development	432	147	401	422	9570	Male	Research Scient
Sales	43	14	38	39	827	Male	Manager
Sales	491	159	431	425	10228	Male	Sales Executive
Sales	90	38	101	105	2370	Male	Sales Represent
Human Resources	34	12	35	33	700	Female	Human Resourc
Human Resources	10	3	10	9	93	Female	Manager
Research & Development	125	42	115	118	2732	Female	Healthcare Rep
Research & Development	191	68	188	178	4291	Female	Laboratory Tech
Research & Development	53	17	49	47	1020	Female	Manager
Research & Development	163	54	157	146	3542	Female	Manufacturing
Research & Development	82	25	54	68	1617	Female	Research Direct
Research & Development	273	94	253	251	6626	Female	Research Scient
Sales	46	14	36	35	1005	Female	Manager
Sales	295	102	278	274	6776	Female	Sales Executive
Sales	93	33	95	82	2156	Female	Sales Represent
Total	**3460**	**1190**	**3245**	**3238**	**78308**		

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Department	Sum of Education	EducationField	BusinessTravel	Sum of Attrition Count	Sum of EmployeeCount	Sum of EnvironmentSatisfaction	Gen
Human Resources	7	Human Resources	Non-Travel	0	2	7	Fem
Human Resources	6	Human Resources	Non-Travel	0	2	4	Mal
Human Resources	6	Human Resources	Travel_Frequently	2	2	5	Fem
Human Resources	10	Human Resources	Travel_Frequently	0	3	6	Mal
Human Resources	6	Human Resources	Travel_Rarely	0	2	6	Fem
Human Resources	34	Human Resources	Travel_Rarely	3	11	31	Mal
Human Resources	4	Life Sciences	Travel_Frequently	1	1	2	Fem
Human Resources	3	Life Sciences	Travel_Frequently	0	1	3	Mal
Human Resources	17	Life Sciences	Travel_Rarely	0	6	17	Fem
Human Resources	13	Life Sciences	Travel_Rarely	0	4	11	Mal
Human Resources	5	Medical	Non-Travel	0	1	4	Mal
Human Resources	6	Medical	Travel_Frequently	0	2	2	Mal
Human Resources	4	Medical	Travel_Rarely	1	2	8	Fem
Human Resources	16	Medical	Travel_Rarely	1	6	14	Mal
Human Resources	1	Medical	TravelRarely	0	1	4	Mal
Human Resources	4	Other	Travel_Frequently	0	1	4	Mal
Human Resources	5	Other	Travel_Rarely	0	2	5	Mal
Human Resources	7	Technical Degree	Travel_Rarely	1	3	7	Mal
Research & Development	25	Life Sciences	Non-Travel	1	8	22	Fem
Research & Development	71	Life Sciences	Non-Travel	1	27	73	Mal
Research & Development	91	Life Sciences	Travel_Frequently	7	32	95	Fem
Research & Development	127	Life Sciences	Travel_Frequently	12	46	112	Mal
Research & Development	266	Life Sciences	Travel_Rarely	12	89	229	Fem
Research & Development	408	Life Sciences	Travel_Rarely	16	139	384	Mal
Total	**3460**			**189**	**1190**	**3245**	

6. Business Recommendations

● **Focus on High-Attrition Departments:**
Departments showing higher attrition rates should be investigated to identify root causes such as workload, management issues, or lack of growth opportunities.

● **Improve Work-Life Balance Programs:**
Employees with lower WorkLifeBalance scores should be prioritized for flexible work policies, wellness programs, and workload optimization to reduce burnout and turnover risk.

● **Salary Benchmarking for Key Roles:**
Job roles with below-average MonthlyIncome should be reviewed against industry standards to ensure competitive compensation and prevent loss of top talent.

● **Strengthen Manager Engagement:**
Since YearsWithCurrManager impacts employee stability, HR should promote regular manager–employee check-ins, mentorship programs, and leadership training.

● **Targeted Retention Strategy for High Performers:**
High-income and experienced employees should be given retention bonuses, career path planning, and recognition programs to reduce voluntary attrition.

● **Early Warning Attrition Monitoring:**
Develop dashboards in Power BI to continuously monitor attrition indicators such as tenure, income level, and work-life balance for proactive intervention.

● **Employee Segmentation-Based HR Policies:**
Different strategies should be designed for New, Mid-Level, and Experienced employees to address their unique needs and career expectations.

● **Data-Driven HR Decision Making:**
HR teams should regularly use analytics and SQL reports to support hiring, promotion, and retention decisions instead of relying only on intuition.